Sampling Methods & Bias

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Identifying sampling methods**

A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

This is a simple random sample (SRS): every student has an equal chance of being selected, and every group of 40 students has an equal chance of being the chosen sample.

2 **[Foundational] • Target: Identifying sampling methods**

A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.

This is stratified random sampling. It might be preferred because it guarantees representation from every grade level, which is useful if opinions or characteristics vary meaningfully by grade — an SRS of the whole school could, by chance, under-represent or over-represent a particular grade.

3 **[Foundational] • Target: Cluster sampling and systematic sampling**

A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

This is cluster sampling: entire naturally-occurring groups (homerooms) are randomly selected, and every member of the selected groups is surveyed. This differs from stratified sampling, where the researcher samples from every group, rather than sampling only a few whole groups and surveying everyone within them.

4 **[Foundational] • Target: Cluster sampling and systematic sampling**

A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

This is systematic sampling: individuals are selected at a fixed interval (every 25th item) from an ordered list or sequence, typically starting from a randomly chosen point.

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5 [Foundational] • Target: Sources of bias

A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

This is voluntary response bias: students with strong opinions (especially strongly negative or strongly positive ones) are more likely to take the time to respond, so the 12% who responded are not representative of the full student body's range of opinions — moderate or indifferent students are underrepresented.

6 [Foundational] • Target: Sources of bias

A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.

This introduces undercoverage bias: households that only use cell phones (increasingly common, especially among younger people) are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

7 [Foundational] • Target: Wording and response bias

A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

The question is leading — the word "wasteful" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results. A neutral rewrite: "Do you support or oppose the new cafeteria policy?"

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Wording and response bias

Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

This is response bias: when asked sensitive questions directly by an interviewer, respondents may feel social pressure to give a more socially acceptable answer rather than the truthful one, especially without the anonymity that a written or online survey might provide.

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9 **[Intermediate]** • **Target: Comparing sample size and bias**

Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

Bias is a systematic flaw in how the sample was selected, not a matter of the sample being too small — collecting responses from a larger number of self-selected volunteers would still overrepresent people with strong opinions. Only fixing the sampling method itself (e.g. switching to a simple random sample of all students) removes the bias; a larger biased sample is still biased, just with a more precise estimate of the wrong quantity.

10 **[Intermediate]** • **Target: Designing a sampling plan**

A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.

Stratified random sampling is recommended: divide the population into the 4 neighborhoods (strata), then take a simple random sample of residents from each neighborhood, proportional to (or intentionally balanced across) each neighborhood's population. This ensures every income group is represented, rather than relying on chance to include them proportionally as an SRS of the whole city might not.

11 **[Intermediate]** • **Target: Convenience sampling and its pitfalls**

A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

This is convenience sampling: the researcher selects whoever is easiest to reach rather than using any random selection process. The results cannot be generalized to the whole city because people at a gym are not representative of the general population — they are systematically more likely to exercise regularly than the average city resident, biasing the sample toward higher reported exercise habits.

12 **[Intermediate]** • **Target: Comparing sampling methods side by side**

A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

Cluster sampling: randomly select 20 stores, then survey all customers at those 20 stores — cheaper and more logistically convenient since fieldwork is concentrated in fewer physical locations. Stratified sampling: divide the 500 stores into regions (strata), then sample customers from every region — guarantees geographic representation across all regions rather than relying on the 20 randomly chosen stores to happen to cover every region well.

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13 [Intermediate] • Target: Identifying the population and sample

A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

Population: all 30,000 enrolled students at the university. Sample: the 400 students who were randomly selected and surveyed.

14 [Intermediate] • Target: Identifying the population and sample

A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

This scenario has at least two compounding sources of bias: (1) undercoverage — only viewers of that specific broadcast, who own a phone capable of texting and chose to watch that program, are even eligible to respond, excluding large portions of the public; (2) voluntary response bias — among those eligible, only viewers with strong opinions (often skewed toward one side) are motivated to actually text in. Together, these make the reported percentages reflect the opinions of a self-selected, unrepresentative subgr.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

Step 1: treat each of the 12 departments as a separate stratum. Step 2: take an independent simple random sample of patients from within each department. Step 3: combine the department-level samples into the overall estimate (optionally weighting by each department's size). This guarantees every department is represented and allows department-by-department comparisons. A simple random sample of all patients, by contrast, might by chance draw very few patients from a smaller department, making it hard to estimate or.

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16 **[Challenging] • Target: Identifying the population and sample • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.

A cluster sample would involve randomly selecting a small number of entire coffee shops (say, 15 out of 200) and measuring service time for every customer at those 15 locations. A key risk is that if certain regions or store types happen to differ meaningfully in typical service time, the randomly selected 15 shops could, purely by chance, fail to represent that variation well (e.g. all 15 could happen to be busy urban locations). A stratified sample based on region would guarantee representation from every region.

17 **[Challenging] • Target: Comparing sampling methods side by side • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

(a) Stratified Sampling: Guarantees representation from every high school. Because schools may have differing cultures and student demographics (homogeneous within schools, heterogeneous between schools), stratifying reduces sampling variability between strata, providing a more precise district-wide estimate. (b) Cluster Sampling: Highly practical and cost-effective because surveyors only visit 2 physical campuses. However, if individual schools differ substantially from one another, cluster sampling introduces muc.

18 **[Challenging] • Target: Identifying the population and sample • Cross-Lesson: 1.01 Measures of Center & Spread**

Level 3 · AdvancedA university alumni association randomly selects 500 graduates from its directory and sends them a survey asking for their current annual salary. Only 95 completed surveys are returned, yielding an average reported salary of 115,000. (a) Name the primary source of bias present. (b) Explain the likely direction of the bias on the estimated mean salary. (c) Explain whether the results can be generalized to all graduates of the university.

(a) Source of Bias: Nonresponse bias. A representative random sample was selected, but 81% (405 of 500) of selected individuals failed to respond. (b) Direction of Bias: The estimate is likely an overestimate (upward bias). Graduates with high incomes and prestigious careers are substantially more eager and proud to report their earnings, while graduates who are unemployed or earn lower wages are far less likely to respond. (c) Generalizability: No, the results cannot be generalized to all graduates because the mas.

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19 [Challenging] • Target: Identifying sampling methods • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Level 1 · FoundationsA random sample of 100 high school seniors finds that 64 plan to attend a four-year college. Construct a 95% confidence interval for the true population proportion p of seniors attending a four-year college.

Point estimate: $\hat{p} = \frac{64}{100} = 0.64$. Critical value: $z^* = 1.96$. Standard error: $\text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.64(0.36)}{100}} = \sqrt{\frac{0.2304}{100}} = 0.048$. Margin of error: $\text{ME} = 1.96 \times 0.048 = 0.09408 \approx 0.094$. Confidence interval: $0.64 \pm 0.094 = \mathbf{(0.546, 0.734)}$.

20 [Challenging] • Target: Cluster sampling and systematic sampling • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

$142 \pm 2.576 \cdot \frac{20}{\sqrt{100}} = 142 \pm 5.15 = (136.85,\ 147.15)$. "99% confidence" means: if this process were repeated many times, 99% of the intervals produced would contain the true μ.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying sampling methods

A factory assigns each worker a number from 1 to 1223, then uses a random number generator to select 70 numbers, surveying those workers. Identify this sampling method.

a. Convenience sample

b. **Simple random sample (SRS) ✓**

c. Stratified random sample

d. Cluster sample

Simple random sample (SRS). Every worker has an equal chance of being selected, and every possible group of 70 workers has an equal chance of being chosen — this is the defining feature of an SRS.

22 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher divides a factory into groups (morning shift, afternoon shift, and night shift), then takes a simple random sample of 53 workers from each group. Identify this sampling method.

a. Cluster sample

b. Convenience sample

c. **Stratified random sample ✓**

d. Systematic sample

Stratified random sample. The population is divided into groups (strata), and an independent simple random sample is taken from every stratum — this is stratified random sampling.

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23 [Foundational] • Target: Sources of bias

A researcher randomly selects 3 entire delivery routes out of 33 at a shipping company, and surveys every member within those selected delivery routes. Identify this sampling method.

a. Stratified random sample

b. Simple random sample (SRS)

c. Systematic sample

d. **Cluster sample ✓**

Cluster sample. Whole naturally-occurring groups (delivery routes) are randomly selected, and every individual within a selected group is surveyed — this is cluster sampling.

24 [Foundational] • Target: Wording and response bias

A quality-control inspector at a factory checks every 49th item coming off a production line, starting from a randomly chosen point among the first 49 items. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. **Systematic sample ✓**

d. Stratified random sample

Systematic sample. Individuals are selected at a fixed interval (every 49th item) from an ordered sequence, after a random starting point — this is systematic sampling.

25 [Foundational] • Target: Comparing sample size and bias

A researcher stands outside a farmers market and surveys the first 44 people encountered, intending to represent the general population. Identify this sampling method.

a. Systematic sample

b. Cluster sample

c. Simple random sample (SRS)

d. **Convenience sample ✓**

Convenience sample. The researcher selects whoever is easiest to reach rather than using any random selection process — this is convenience sampling.

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26 [Foundational] • Target: Designing a sampling plan

A company assigns each employee a number from 1 to 1017, then uses a random number generator to select 61 numbers, surveying those employees. Identify this sampling method.

a. Simple random sample (SRS) ✓

b. Convenience sample

c. Cluster sample

d. Systematic sample

Simple random sample (SRS). Every employee has an equal chance of being selected, and every possible group of 61 employees has an equal chance of being chosen — this is the defining feature of an SRS.

27 [Foundational] • Target: Convenience sampling and its pitfalls

A researcher divides a hospital into groups (department A, department B, and department C), then takes a simple random sample of 72 patients from each group. Identify this sampling method.

a. Cluster sample

b. Stratified random sample ✓

c. Convenience sample

d. Simple random sample (SRS)

Stratified random sample. The population is divided into groups (strata), and an independent simple random sample is taken from every stratum — this is stratified random sampling.

28 [Foundational] • Target: Comparing sampling methods side by side

A researcher randomly selects 6 entire wards out of 18 at a hospital, and surveys every member within those selected wards. Identify this sampling method.

a. Stratified random sample

b. Convenience sample

c. Systematic sample

d. Cluster sample ✓

Cluster sample. Whole naturally-occurring groups (wards) are randomly selected, and every individual within a selected group is surveyed — this is cluster sampling.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Identifying the population and sample

A researcher stands outside a train station and surveys the first 49 people encountered, intending to represent the general population. Identify this sampling method.

a. Cluster sample

b. Simple random sample (SRS)

c. **Convenience sample ✓**

d. Systematic sample

Convenience sample. The researcher selects whoever is easiest to reach rather than using any random selection process — this is convenience sampling.

30 [Intermediate] • Target: Identifying sampling methods

A hospital assigns each patient a number from 1 to 1075, then uses a random number generator to select 80 numbers, surveying those patients. Identify this sampling method.

a. Systematic sample

b. **Simple random sample (SRS) ✓**

c. Stratified random sample

d. Cluster sample

Simple random sample (SRS). Every patient has an equal chance of being selected, and every possible group of 80 patients has an equal chance of being chosen — this is the defining feature of an SRS.

31 [Intermediate] • Target: Cluster sampling and systematic sampling

A researcher divides a factory into groups (freshmen, sophomores, juniors, and seniors), then takes a simple random sample of 67 workers from each group. Identify this sampling method.

a. Cluster sample

b. Systematic sample

c. **Stratified random sample ✓**

d. Simple random sample (SRS)

Stratified random sample. The population is divided into groups (strata), and an independent simple random sample is taken from every stratum — this is stratified random sampling.

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32 [Intermediate] • Target: Sources of bias

A researcher randomly selects 9 entire homeroom classes out of 15 at a school, and surveys every member within those selected homeroom classes. Identify this sampling method.

a. Simple random sample (SRS)

b. Systematic sample

c. **Cluster sample ✓**

d. Convenience sample

Cluster sample. Whole naturally-occurring groups (homeroom classes) are randomly selected, and every individual within a selected group is surveyed — this is cluster sampling.

33 [Intermediate] • Target: Wording and response bias

A quality-control inspector at a clinic checks every 32nd item coming off a production line, starting from a randomly chosen point among the first 32 items. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. **Systematic sample ✓**

d. Cluster sample

Systematic sample. Individuals are selected at a fixed interval (every 32nd item) from an ordered sequence, after a random starting point — this is systematic sampling.

34 [Intermediate] • Target: Comparing sample size and bias

A survey about a new dress code is voluntarily filled out online by residents, and only 20% respond. Identify the type of bias most likely present.

a. Wording bias (leading question)

b. Convenience sampling bias

c. **Voluntary response bias ✓**

d. Response bias (social desirability)

Voluntary response bias. Residents with strong opinions are far more likely to take the time to respond, so the 20% who responded are not representative of the full population's range of opinions.

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35 [Intermediate] • Target: Designing a sampling plan

A phone survey about university services is conducted using only landline phone numbers, systematically excluding residents without internet access. Identify the type of bias this introduces.

a. Voluntary response bias

b. Response bias (social desirability)

c. **Undercoverage bias ✓**

d. Wording bias (leading question)

Undercoverage bias. Residents without internet access are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

36 [Intermediate] • Target: Convenience sampling and its pitfalls

Respondents are asked directly, face-to-face, about underage drinking. Identify the type of bias likely to affect their answers.

a. Wording bias (leading question)

b. Voluntary response bias

c. Convenience sampling bias

d. **Response bias (social desirability) ✓**

Response bias (social desirability). When asked sensitive questions directly by an interviewer, respondents often give a more socially acceptable answer rather than the truthful one.

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37 [Intermediate] • Target: Comparing sampling methods side by side

A survey question reads: "Don't you agree that the wasteful dress code should be changed?" Identify the type of bias in this question.

a. Convenience sampling bias

b. Voluntary response bias

c. **Wording bias (leading question) ✓**

d. Response bias (social desirability)

Wording bias (leading question). The word "wasteful" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 1.01 Measures of Center & Spread

A researcher surveys people at a farmers market and generalizes the results to the entire city's population. Identify the type of bias most likely present.

a. Response bias (social desirability)

b. **Convenience sampling bias ✓**

c. Voluntary response bias

d. Undercoverage bias

Convenience sampling bias. People found at a farmers market are not representative of the general population, since the location systematically attracts a particular type of person.

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39 [Challenging] • Target: Identifying sampling methods • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A survey about a new policy is voluntarily filled out online by employees, and only 14% respond. Identify the type of bias most likely present.

a. Convenience sampling bias

b. Undercoverage bias

c. **Voluntary response bias ✓**

d. Response bias (social desirability)

Voluntary response bias. Employees with strong opinions are far more likely to take the time to respond, so the 14% who responded are not representative of the full population's range of opinions.

40 [Challenging] • Target: Cluster sampling and systematic sampling • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A phone survey about university services is conducted using only landline phone numbers, systematically excluding people who don't check email. Identify the type of bias this introduces.

a. Response bias (social desirability)

b. Convenience sampling bias

c. Voluntary response bias

d. **Undercoverage bias ✓**

Undercoverage bias. People who don't check email are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

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41 [Challenging] • Target: Sources of bias • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Respondents are asked directly, face-to-face, about tax evasion. Identify the type of bias likely to affect their answers.

a. Undercoverage bias

b. **Response bias (social desirability) ✓**

c. Wording bias (leading question)

d. Convenience sampling bias

Response bias (social desirability). When asked sensitive questions directly by an interviewer, respondents often give a more socially acceptable answer rather than the truthful one.

42 [Challenging] • Target: Wording and response bias • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A survey question reads: "Don't you agree that the dangerous parking policy should be changed?" Identify the type of bias in this question.

a. Undercoverage bias

b. Voluntary response bias

c. **Wording bias (leading question) ✓**

d. Response bias (social desirability)

Wording bias (leading question). The word "dangerous" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results.

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43 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher surveys people at a shopping mall entrance and generalizes the results to the entire city's population. Identify the type of bias most likely present.

a. Undercoverage bias

b. Response bias (social desirability)

c. Wording bias (leading question)

d. **Convenience sampling bias ✓**

Convenience sampling bias. People found at a shopping mall entrance are not representative of the general population, since the location systematically attracts a particular type of person.

44 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 3.03 The Normal Distribution

A survey about a proposed schedule change is voluntarily filled out online by employees, and only 8% respond. Identify the type of bias most likely present.

a. **Voluntary response bias ✓**

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

Voluntary response bias. Employees with strong opinions are far more likely to take the time to respond, so the 8% who responded are not representative of the full population's range of opinions.

45 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A phone survey about school services is conducted using only landline phone numbers, systematically excluding cell-phone-only households. Identify the type of bias this introduces.

a. Convenience sampling bias

b. Wording bias (leading question)

c. **Undercoverage bias ✓**

d. Voluntary response bias

Undercoverage bias. Cell-phone-only households are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

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